

PHYSICS 101 · RECITATION HOMEWORK

Homework 2: Light

Name: _____

Date: _____

Instructions: About 20 minutes. Part C is where the week's idea lives: explain the pulse ox in your own words. The flashlight lab at the end needs a dark room. Total: **21 points**.

Part A — The rainbow

1. Write the colors of the rainbow in order from the **LONGEST** waves to the **SHORTEST**. Then name the invisible color just past each end. (3 pts)

2. Draw Newton's experiment: a beam of white light entering a prism and fanning out into colors on the other side. Label the red end and the violet end. (2 pts)

Part B — Why things have color

3. For each object, write which color(s) bounce back to your eye and what happens to the rest. (4 pts)

Object	Bounces back	The rest of the colors...
A ripe tomato		
A green leaf		
White paper		
A black T-shirt		

Part C — The two-color trick

4. In your own words, explain how the pulse oximeter turns two colors of light into an oxygen percentage. Use the words red, infrared, and hemoglobin. (3 pts)

5. Why does the machine use only the PULSING part of the light signal? (1 pt)

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- A. Only the fresh blood in the arteries pulses; skin, bone, and vein blood don't.
- B. The pulsing part is brighter, so it's easier for the detector to see.
- C. Pulsing light doesn't hurt a baby's eyes the way steady light does.
- D. The pulses are the baby's breaths, which is what the machine measures.

Part D — The lamp that cures

6. Why blue? Explain what blue light does to a bilirubin molecule and why that helps a baby whose liver isn't ready. (2 pts)

7. What did the nurse Sister Jean Ward notice in 1956 that started it all? (1 pt)

Part E — Flashbacks

8. From NICU Tutor: why are preemies on extra oxygen kept around 90-95% saturation instead of 100%? (1 pt)

9. From last week's Physics: a CPAP is set at 6 cmH₂O. About how many mmHg is that? (1 mmHg ~ 1.36 cmH₂O.) (1 pt)

Part F — Think like a nurse

10. A preemie's pulse ox suddenly reads 75%. The baby is pink, wiggling, and her heart rate on the monitor is a steady 145. You look at her foot: the band has slid sideways and its little red light is shining onto the mattress. Explain, using today's physics, why the number is wrong — and what you'd do first, second, and third. (3 pts)

HOME LAB (optional, not graded): Your finger is a pulse oximeter

You need: a small flashlight or phone light, a dark room, and (bonus) scraps of red and blue cellophane or candy wrappers

1. In the dark, press the flashlight against the pad of your fingertip so it shines INTO the finger.
2. Look at the other side of the finger. What color is the glow? White went in — what came out?
3. Bonus: hold red cellophane over the light and try again; then blue. Which color gets through your finger?
4. Bonus 2: tilt a CD or DVD under a lamp and find the rainbow. Which color is on the outside edge?

What I saw:

Done! Professor's initials: _____

Score: _____ / 21 points

Professor's comment:
